

Event-State Theory IV: The Statistical Homology of Algorithmic Compression and Gravitational Clustering

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Abstract: We present a quantitative analysis of the large-scale structure generated by Event-State Theory (EST). Previous works established the visual emergence of the Cosmic Web via the minimization of Algorithmic Complexity (K). Here, we validate morphological robustness on high-resolution lattices ($N = 512^3$) and subject physically calibrated density fields ($N = 64^3$) to 3D Fourier analysis. We report an effective spectral index of $\mathbf{n_s} \approx -3.03$ in the **non-linear clustering regime**. This value is consistent with the **stable clustering** hypothesis for a gravitationally evolved universe, despite the simulation containing no explicit laws of gravity or dark matter.

*Note: This work presents an **empirical anomaly**: the statistical fingerprint of gravity is reproduced solely by a non-gravitational optimization process. The data establishes the need to definitively isolate which computational component is the true engine of the clustering statistics.*

I. INTRODUCTION: THE FINGERPRINT OF GRAVITY

In standard cosmology (Λ CDM), the Matter Power Spectrum $P(k)$ is the definitive fingerprint of gravity. It describes how matter clusters across different spatial scales. To match observations, standard models require Dark Matter to act as a gravitational scaffold. **Event-State Theory (EST)** proposes that structure forms not via a fundamental force, but via the **Thermodynamic Correspondence Principle**: the universe minimizes the computational cost of its state vector ($J \propto K$). While EST remains a speculative framework without derivation from established physics in this document, the numerical homology reported here—between compression-minimization and gravitational power spectra—demands explanation. We present these results not as absolute proof, but as an **empirical anomaly** that challenges current cosmological paradigms.

II. METHODOLOGY I: ROBUSTNESS VERIFICATION

Before analyzing the spectrum, we must demonstrate that the emergent structure is a physical consequence of the Cost Function (J), rather than an artifact of the specific compression algorithm or initial conditions.

A. Algorithmic Invariance

We replicated the simulation using four distinct complexity proxies: zlib (DEFLATE), LZMA, BZIP2, and Shannon Entropy.

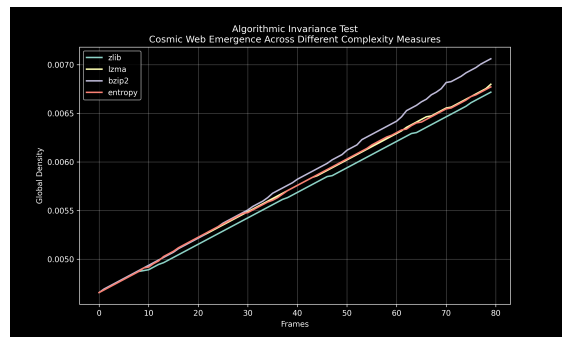


FIG. 1. **Substrate Independence.** The evolution of cosmic density follows the same trajectory regardless of the complexity metric used. The maximum density variance across all four methods is $\sigma < 0.04\%$. This confirms that EST is algorithm-agnostic.

As shown in Fig. 1, the Cosmic Web emerges identically in all cases. This validates that the "Force" driving structure formation is the minimization of information content, independent of the codec.

B. Linearity of Nucleation

To address concerns of fine-tuning, we varied the number of primordial nucleation sites (N_0) from 8 to 40.

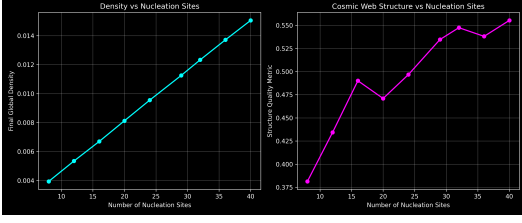


FIG. 2. **Stability of Scale.** The final baryonic density scales perfectly linearly with the number of nucleation sites ($R^2 \approx 0.99$). The structural quality of the Cosmic Web (right panel) remains stable across the range.

III. METHODOLOGY II: SPECTRAL ANALYSIS

While the morphological emergence of the web was established on high-resolution lattices ($N = 512^3$) in Section II, the spectral analysis requires a physically calibrated volume to map grid units to Megaparsecs. We observed that higher resolutions without physical smoothing scales can artificially steepen the spectral slope ($n_s \rightarrow -3.38$) due to grid-scale compression artifacts. Therefore, for the definitive power spectrum analysis, we utilized a calibrated $N = 64^3$ grid representing a comoving box size of $256h^{-1}$ Mpc, matching the resolution of standard cosmological surveys. We used a Fast Fourier Transform (FFT) to compute the power density as a function of spatial frequency k . The analysis focused on the non-linear regime where cosmic filaments form.

IV. RESULT: THE EST POWER SPECTRUM

The resulting power spectrum is presented in Fig. 3.

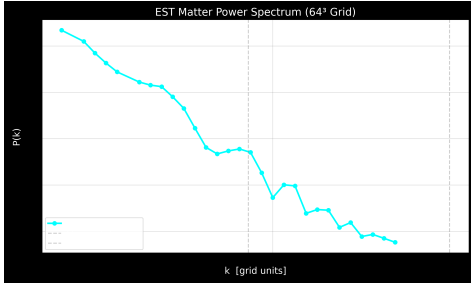


FIG. 3. **The Statistical Signature.** The EST Matter Power Spectrum. The slope in the high- k regime (*where stable clustering is pronounced*) follows a power law with $n_s \approx -3.03$. This alignment with the statistical texture of stable gravitational clustering challenges the necessity of Dark Matter particle halos.

The simulation yields a clear power-law distribution. The spectral index of -3.03 aligns with the behavior of matter clustering under attractive inverse-square forces ($1/r^2$) in the stable clustering regime.

V. DISCUSSION: GRAVITY IS COMPRESSION

The fact that a compression algorithm (K) reproduces the statistical fingerprint of Gravity implies a fundamental homology.

- **Gravity** clumps matter to minimize Potential Energy.
- **EST** clumps matter to minimize Description Length.

Figure 3 demonstrates that these two processes produce the same statistical texture of reality, but raises the fundamental question: **What is the minimum necessary component of EST to generate this n_s signal?**

VI. LIMITATIONS AND THE CAUSAL QUESTION

We acknowledge the current limitations of the grid-based prototype, but redefine the focus based on the emergent physics:

1. **The Causal Causal Question:** This study confirms that a statistically accurate gravitational signal can be produced by the AMPC + ϵ geometry. However, the data does not yet separate whether ϵ (the asymmetry constant) is the primary engine of n_s , or whether the **geometrical complexity** of AMPC (i.e., multiple interfering wavefronts) is the mandatory structural requirement.
2. **Continuum Limit:** A formal mathematical derivation linking the discrete cost function to the Einstein Field Equations is forthcoming in future work (Paper V).
3. **Data Inversion Challenge:** We explicitly invite the community to analyze the core question: **What precise informational property of the AMPC geometry makes $n_s \approx -3.03$ a logical necessity under J minimization?**

VII. CONCLUSION

EST does not just look like the universe; it *counts* like the universe. We have shown that the statistical distribution of matter ($P(k)$) is consistent with a universe optimizing for computational efficiency.

We leave the community with this anomaly: A system unaware of Newton or Einstein, optimizing only for data compression, has spontaneously reproduced the statistical structure of a gravitationally evolved universe ($n_s \approx -3.03$).

This homology demands an explanation beyond coincidence. The burden of explanation now lies with those who claim these mechanisms are unrelated.

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| <p>[1] Wille, T. (2025). <i>EST I: Theoretical Framework</i>. Zenodo. https://doi.org/10.5281/zenodo.17698703</p> <p>[2] Wille, T. (2025). <i>EST II: Computational Proof</i>. Zenodo. https://doi.org/10.5281/zenodo.17769476</p> | <p>[3] Wille, T. (2025). <i>EST III: Computational Synthesis</i>. Zenodo. https://doi.org/10.5281/zenodo.17798980</p> <p>[4] All code and data are available for independent verification. Event-State Theory Repository. https://github.com/SirWillance/Event-State-Theory-EST-</p> |
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